

LUSONG

Post-flood leptospirosis early warning and prophylaxis dispatch for every barangay

“Lumusong ka man, may bantay.”

Team TeamLusong | PHINMA Araullo University, Cabanatuan City

Philippine Startup Challenge XI (2026)

Initial Presentation Deck | Sector: Health | Technology: AI / SaaS

Problem Description

After every flood, a preventable epidemic follows two to four weeks later

~12,000

leptospirosis cases in 2026 by Oct 1, 46% above 2025 [6]

4,859

cases in just 13 weeks of the 2025 rainy season (Jun 8 to Sep 4) [5]

3.6 M

people in 9,643 barangays affected by Typhoon Uwan alone, Nov 2025 [2]

6%

of health facilities had internet connectivity in 2024 [12]

- Leptospirosis is caught by wading through floodwater contaminated by rat urine. Severe cases need dialysis; deaths cluster among working-age adults who waded to work, clear drains, or rescue neighbors.
- Prevention exists and is cheap: DOH recommends doxycycline post-exposure prophylaxis for people who waded, most effective within 24 to 72 hours.
- The gap: nobody knows, in time, which households waded. Warnings are generic, prophylaxis arrives late, and surveillance counts cases only after they reach hospitals (CDC EID, Mar 2026; PMC 2025) [7][8].
- The frontline is thin and offline: 21.2 health workers per 10,000 vs WHO 44.5; 26,504 barangay health stations for 42,026 barangays [9] [11].

Why Existing Tools Do Not Solve It

Hazard platforms are excellent; the post-flood health response layer is empty

Capability	HazardHunterPH / GeoRiskPH	PAGASA + Google flood forecasts	DOH advisories	BIMS / eLGU	Telemedicine apps	LUSONG
Pre-event hazard maps	Yes	Yes	No	No	No	Uses them
Finds exposed households after a flood	No	No	No	No	No	Yes
Targets prophylaxis within 72 h	No	No	Generic text	No	No	Yes
BHW house-to-house workflow, offline	No	No	No	Records only	No	Yes
Symptom screening and referral	No	No	No	No	Online, urban	Yes, offline, rural
Measures outcome per barangay	No	No	No	No	No	Yes

DILG mandated the Barangay Information Management System (BIMS) for all barangays in Dec 2025, with 25,000+ barangays oriented by mid-2026 [15]. Household-level data now exists almost everywhere. Nobody has used it for post-flood health response yet.

Sustainable Development Goal (SDG) Priority

Primary: SDG 3 Good Health and Well-being

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Target 3.3: end the epidemics of water-borne and other communicable diseases

Target 3.d: strengthen capacity for early warning, risk reduction and management of national and global health risks

Secondary SDGs

- SDG 11.5: reduce deaths and the number of people affected by disasters, including water-related disasters
- SDG 13.1: strengthen resilience and adaptive capacity to climate-related hazards
- SDG 6.2: sanitation, through hotspot mapping of flood and waste exposure
- Alignment: Universal Health Care Act (interoperable records by 2028), RA 10121 (70% of LDRRMF for prevention and preparedness), ASEAN 2026 theme “Navigating Our Future, Together” (digital public goods, climate resilience)

Target Market Segment

Beneficiaries, users and the customers who pay

Segment	Who	Size	Role
Beneficiaries	Residents of flood-prone barangays; working-age adults who wade most	8.2 M people affected by floods and cyclones in 2025; 9,643 barangays hit by Uwan alone [2][3]	Receive advisories and prophylaxis
Primary users	Barangay Health Workers, RHU nurses and physicians, MDRRMO officers	42,026 barangays; 26,504 barangay health stations [11]	Run the house-to-house response
Paying customers	City and municipal governments (LDRRMF prevention funds, health budget)	About 1,640 LGUs; est. 600 municipalities + 60 cities with recurring floods	Annual subscription
Early adopters	Marikina, Malabon, Navotas, Calumpit, Hagonoy, Cabanatuan, Dagupan, Naga, Ormoc, Cotabato City, Davao riverside districts	Active MDRRMO, BIMS-oriented barangays, eLGU accounts (41 cities, 901 municipalities) [15]	Pilot and reference customers

Solution

LUSONG turns a flood event into a barangay-level health action list within hours

LUSONG Command

LGU and RHU web dashboard

- Exposure engine fuses rainfall, flood depth maps, elevation, and BIMS household locations
- Ranks barangays by expected cases (AI model trained on past rainfall, flood extent and DOH weekly case data)
- Doxycycline need vs stock per barangay health station

LUSONG Bantay

Offline-first BHW mobile app

- Prioritized house-to-house list per purok
- 3-question Filipino symptom screen (fever, calf pain, red eyes or jaundice)
- Prophylaxis log and one-tap RHU referral; syncs when signal returns

LUSONG Abiso

Resident messaging

- Targeted SMS in Filipino and regional languages within hours of a flood
- Residents reply 1 or 2 to self-report wading and symptoms
- Messenger and eGovPH channels next

How It Works

No hardware required; built on data the government already publishes

1. Trigger

PAGASA rainfall, flood forecast, or MDRRMO report

2. Exposure

Spatial join of flood layers with BIMS households (PostGIS)

3. Risk ranking

Gradient-boosted model on rainfall, flood extent, past cases

4. Dispatch

SMS to households; BHW lists pushed offline

5. Response

Prophylaxis logged, symptoms screened, referrals made

6. Learn

30-day case tracking; reports in DOH FHSIS format

Stack: Python (FastAPI, XGBoost, GeoPandas), PostgreSQL + PostGIS, React dashboard, progressive web app with IndexedDB offline sync, local SMS gateway, small language model for multilingual advisories with RHU sign-off.

Features

Must-have features ship in the 8-week MVP; the rest follow after the first pilot

MVP (must have)

- Barangay map with household exposure scoring
- BIMS or census CSV import (no re-encoding)
- SMS blast with reply capture (1 = waded, 2 = symptoms)
- Offline BHW checklist and symptom screen
- Doxycycline stock vs need panel
- Referral queue for the RHU
- Exportable report in DOH FHSIS format

Next (after pilot)

- ML risk ranking trained on DOH weekly surveillance
- Messenger and eGovPH super app channels
- Rain gauge and community sensor ingestion
- Dengue, acute diarrhea and heat-illness modules
- Province and DOH regional dashboards
- GeoRiskPH API integration under MOU

The 75-Second Demo

What judges will see on stage

1. Pick a barangay; simulate 180 mm rainfall and 0.5 m flood depth
2. Map turns red: 1,240 exposed households, 2,900 people
3. Doxycycline need 2,900 doses vs 400 in stock; reorder suggested
4. One tap: the judge's phone receives the Tagalog SMS
5. BHW tablet, offline, shows the house-to-house list
6. A symptom report escalates to the RHU referral queue

Sample SMS

LUSONG: Lumusong ka ba sa baha kahapon? Pumunta sa Barangay Health Station para sa libreng doxycycline ngayong araw. Kung may lagnat o sakit ng binti sa susunod na 2 linggo, magpakonsulta agad. Reply 1 kung lumusong, 2 kung may sintomas.

Business Model

LGUs pay from the prevention share of the Local DRRM Fund; prevention is cheaper than dialysis

Revenue stream	Price (PHP, to be validated)
Municipal subscription (up to 30 barangays)	60,000 per year + 1,500 per extra barangay
Component city subscription	150,000 to 300,000 per year
Implementation and BHW training	25,000 one-time per LGU
SMS credits	Cost plus 20%
Province or DOH regional license	Volume pricing
Future: anonymized outbreak analytics	Insurers, pharma distributors (DPA-compliant)

Unit economics (estimate)

- Cloud and SMS cost per municipality about PHP 12,000 per year
- Onboarding about 3 person-days
- Gross margin above 70% after year one
- 10 LGUs cover the team's operating cost; 50 LGUs fund a full-time team of five
- Procurement through LDRRMF prevention allocation (RA 10121) is annual and renewable
- One avoided dialysis case costs more than a year of LUSONG

Market Size and Demand

Philippine LGU market first; exportable to ASEAN neighbors with endemic leptospirosis

PHP 131 M / yr

TAM: about 1,640 cities and municipalities x PHP 80,000 average (estimate); PHP 180 M with province and DOH licenses

PHP 48 M / yr

SAM: est. 600 flood-prone municipalities x 60,000 + 60 cities x 200,000 (Uwan alone hit 725 LGUs [2])

PHP 8 M / yr

SOM in 3 years: 100 LGUs plus SMS and training revenue

- Leptospirosis cases up 46% year on year in 2026 [6]
- 2026 GAA restored about PHP 129.8 B to PhilHealth; health sector budget PHP 448 B [16]
- DILG BIMS mandate (Dec 2025) supplies the household data input; UHC Act pushes LGUs toward interoperable digital health tools by 2028 [12][15]
- Philippine ASEAN chairmanship 2026 prioritizes digital public goods and climate resilience; leptospirosis is endemic in Indonesia, Thailand, Malaysia and Vietnam [17]
- No Philippine startup or agency product found that targets leptospirosis exposure and prophylaxis logistics

Progress / Traction

Status as of October 2026: idea stage with a validated problem and a build plan

Done

- Problem sized from DOH surveillance, NDRRMC and CDC/PMC literature (2025 to 2026)
- Competitive scan: hazard platforms, telemedicine, govtech; no post-flood health response product found
- Solution architecture, offline-first BHW workflow and multilingual SMS designed
- Exposure engine data sources identified: PAGASA, GeoRiskPH, NAMRIA, BIMS exports, DOH weekly tables

Next 8 weeks (commitments)

- 15 interviews: 5 BHWs, 3 RHU clinicians, 3 MDRRMO officers, 2 barangay captains, 2 patient families
- Backtest the exposure engine on two 2024 to 2025 flood events against DOH weekly data
- Letter of Intent from one flood-prone LGU; 20 BHWs trained; 1,000 households enrolled
- Live pilot in one or two barangays; publish delivery time and reach numbers

Roadmap

From MVP to ten paying LGUs in twelve months

Milestone	Target date	Measure
MVP with backtested exposure engine	Regional pitch, Nov 2026	>= 70% recall of barangays with later lepto clusters (estimate)
First LGU LOI or MOA, 20 BHWs trained	Dec 2026	1,000 households enrolled in LUSONG Abiso
Live through one rainy-season flood in 3 LGUs	Jun 2027	>= 60% of exposed households reached within 24 h; prophylaxis offer within 72 h
Outcome evaluation vs matched control barangays	Dec 2027	Flood-to-first-dose under 48 h; fewer hospitalized cases
10 paying LGUs, second disease module	Dec 2027	PHP 600,000 annual recurring revenue

Proposed Use of Funds / Asks

PHP 480,000 for twelve months: MVP completion and pilots in three LGUs

Item	PHP
Cloud, maps and model hosting	60,000
SMS and messaging credits (about 150,000 messages)	90,000
Field validation and pilot travel	80,000
BHW training materials and five shared devices	60,000
Multilingual content and clinical review	40,000
Legal, data privacy and registration	30,000
Student team stipends	80,000
Contingency	40,000
Total	480,000

Our asks

- Pilot introductions to two flood-prone LGUs and their municipal health officers
- DICT Startup Grant Fund consideration (PHP 500,000 to 1 M) for year-one pilots
- DOH Epidemiology Bureau and GeoRiskPH data access under MOU
- Mentorship on LGU procurement and Data Privacy Act compliance
- Allocation: 44% product and infrastructure, 38% field validation and adoption, 18% compliance and contingency

Team

PHINMA Araullo University, Cabanatuan City

Justine Tabunan (CEO)

CEO: customer validation, LGU partnerships, pitch, finance. info@astin.ph

John Alvin D. Bondoc (CTO)

CTO: exposure and risk engine, backend, offline PWA. johnalvinbondoc@gmail.com

Aurchie Keem N. Rodriguez (CPO/CMO)

CPO/CMO: BHW experience, multilingual content, training, brand. aurchierodriguez@gmail.com

Gabriel Nikiloz G. Sioson (CIT Faculty)

Mentor: technical review, DOH Region III and LGU introductions. ggsioson.au@phinmaed.com



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Every flood is followed by a preventable epidemic. LUSONG makes sure the PHP 10 antibiotic reaches the right house in time.

Team TeamLusong | PHINMA Araullo University, Cabanatuan City

Contact: info@astin.ph / 0965 274 3311 (Justine Tabunan); ggsioson.au@phinmaed.com (mentor)

Sources: see Concept Note, Section X and reference list